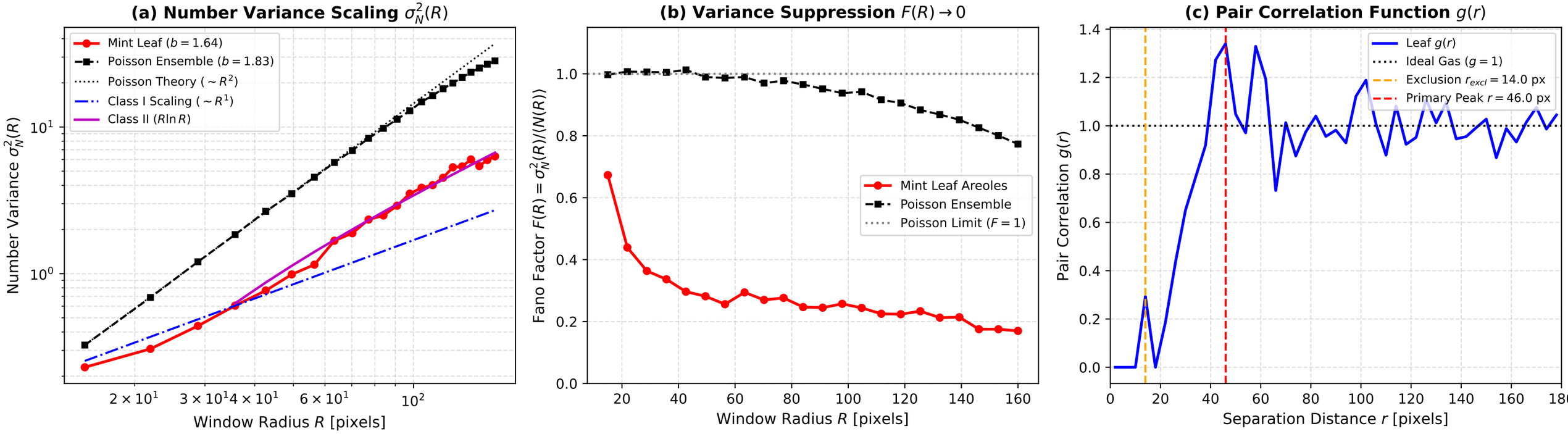
(a) Number Variance Scaling $\sigma_N^2(R)$
Mint Leaf ($b = 1.64$)
Poisson Ensemble ($b = 1.83$)
Poisson Theory ($\sim R^2$)
Class I Scaling ($\sim R^1$)
Class II ($R \ln R$)
Number Variance $\sigma_N^2(R)$
10^1
10^0
2×10^1
3×10^1
4×10^1
6×10^1
10^2
Window Radius R [pixels]
(b) Variance Suppression $F(R) \to 0$
Mint Leaf Areoles
Poisson Ensemble
Poisson Limit ($F = 1$)
Fano Factor $F(R) = \sigma_N^2(R)/\langle N(R) \rangle$
1.0
0.8
0.6
0.4
0.2
0.0
20
40
60
80
100
120
140
160
Window Radius R [pixels]
(c) Pair Correlation Function $g(r)$
Leaf $g(r)$
Ideal Gas ($g = 1$)
Exclusion $r_{excl} = 14.0$ px
Primary Peak $r = 46.0$ px
Pair Correlation $g(r)$
1.4
1.2
1.0
0.8
0.6
0.4
0.2
0.0
0
20
40
60
80
100
120
140
160
180
Separation Distance r [pixels]